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ES327 - Individual Project

Development of Bipolar Pulsed Power Supply for Electrochemical (ECM) Applications

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Abstract

This dissertation investigates the design of a bipolar pulsed power supply for electrochemical machining (ECM) in response to an industrial brief provided by TextureJet Ltd. The target specification required bipolar square-wave operation up to ± 150 V and ± 20 A, with short pulse widths and a maximum operating frequency of 330 kHz. Following a review of ECM pulse requirements and candidate converter topologies, the bipolar high voltage pulsed power source (BHVPPS) was selected as the most suitable basis for development because of its high voltage gain from a low-voltage input, modular structure, and published analytical design procedure. A two-stage version of the converter was then designed and evaluated in LTspice using a time-domain model with ideal switching devices. The simulation reproduced the intended six-mode operating sequence and achieved a pulse-voltage droop of approximately 4.08 %. However, the steady-state output across the resistive load didn't fully reach the target specifications. The study therefore provides an analytical and simulation framework for future hardware development, while also identifying the main technical limitations that must be resolved before practical implementation.

Acknowledgments

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Abbreviations

BBPG Boost Bipolar Pulse Generator

BHVPPS Bipolar High-Voltage Pulsed Power Source

DC Direct Current

DCM Discontinuous Conduction Mode

ECM Electrochemical Machining

EDL Electric Double Layer

IEG Inter-Electrolyte Gap

IGBT Insulated-Gate Bipolar Transistor

MOSFET Metal–Oxide–Semiconductor Field-Effect Transistor

MRR Material Removing Rate

PSU Power Supply Unit

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Chapter 1

Introduction

As of today, a wide range of materials with properties of high strength, hardness, and excellent corrosion resistance are commonly used in industries such as automotive, aerospace, and electronics. However, machining these difficult-to-cut materials presents several challenges, including rapid tool wear, low precision, poor stability, and high costs [1]. Electrochemical machining (ECM) is a non-traditional machining technique that functions by anodic dissolution, which occurs during an electrolytic reaction, and it offers a better alternative in generating accurate 3-D geometries for the difficult-to-machine materials [2]. According to Faraday's law, as a result of separating the anodic workpiece from the cathodic tool at a specified distance, hence, by creating a gap filled with a suitable electrolyte, and then passing a pulsed voltage through the electrolyte, it is possible to deplete and transfer the material through the electrolyte according to the shape of the tool [2]. The non-contact processing nature of ECM enhances its applicability, regardless of the material's hardness or strength, and makes it highly advantageous for numerous machining processes such as deburring, drilling, die sinking, and others, enabling the creation of complex geometrical features [2]. Additional advantages of ECM over traditional machining methods include reduced tool wear, preservation of the material's microstructure, and minimized thermal defects [3]. Moreover, ECM produces components that are stress-free, crack-free, and have bright, smooth surfaces; in fact, ECM is often used for polishing purposes [4].

Modern ECM systems require a precisely configured power supply, which is critical for ensuring a stable machining process based on electric power. The properties of a power

supply have a significant effect on the machining performance metrics such as efficiency, accuracy, and surface quality [1]. Studies report that bipolar pulsed power units delivering short pulses at high frequency significantly enhance ECM performance by improving material removal efficiency, reducing the overcut effect, and providing a more accurate localization of material removal [5] [6]. Ongoing advances in the field of power electronics and high-power switching devices, such as MOSFETs and IGBTs, present new opportunities to optimize ECM power supply design [1]. Despite this, a practical power supply unit (PSU) capable of delivering high current and voltage pulses while remaining compact and cost-effective is largely unavailable, significantly hindering the commercial development and adoption of ECM systems [5]. This study investigates the practical development of a pulsed power supply for ECM, following the specifications provided by Jamie Urquhart-Shirra from TextureJet Ltd.

1.1 Problem Specification

TextureJet Ltd. requires a bipolar pulsed power supply unit for use in their electrochemical machining systems. The project was initially based on a preliminary set of electrical requirements presented in Appendix A. However, as the company progressed in the development of its own power supply architecture, these requirements were later updated. Therefore, the project is carried out following the updated specifications:

- Voltage range: 0 to ± 150 V
- Current range: 0 to ± 20 A
- Waveform: bipolar square wave $\underline{(+ve) V} \rightarrow \underline{(0) V} \rightarrow \underline{(-ve) V} \rightarrow \underline{(0) V} \rightarrow \underline{(+ve) V}$
- Frequency: 330 kHz (Max.)
- Pulse Width: 0.5 μ s (Min.)

1.2 Aim

To design and experimentally validate a safe, scalable, and cost-effective bipolar pulsed PSU for ECM applications that fulfills the technical and operational requirements specified by TextureJet Ltd.

1.3 Objectives

1. Critically review the operating requirements for a bipolar ECM power supply through literature review, analysis of existing ECM systems, and consultation with experts, in order to determine whether the proposed specifications for voltage & current magnitudes, pulse waveform, and pulse timing are appropriate for ECM operation.
2. Evaluate suitable circuit topologies for an ECM power supply capable of delivering the justified requirements.
3. Select and justify the most appropriate power supply circuit topology and switching configurations based on performance, cost, and safety metrics.
4. Design and validate the selected design through simulation using tools such as LT-Spice, verifying its ability to operate under the required ECM load conditions.
5. Assess hardware implementation and provide a detailed framework for future construction and testing of the power supply, including recommended components and control approach.
6. Document and critically assess the project outcomes to determine the level of technical achievement, performance limitations, and to identify opportunities for further improvements and potential commercial implementation of the PSU.

Chapter 2

Literature Review

2.1 Electrical Performance Requirements

Developing a bipolar pulsed PSU for ECM processes requires a careful analysis of the provided electrical performance metrics. The initial requirement for the power supply to generate square waves with voltages up to ± 500 V and currents up to ± 6 A (according to TextureJet) can be classified into two categories: pulse amplitude and pulse waveform & timing. These categories translate specifications into measurable metrics that govern precision, efficiency, and long-term stability of the ECM process, and consequently establish the justified design criteria with which the PSU must be developed. As mentioned earlier, although the project was initially framed around preliminary targets, TextureJet later refined its specifications. Therefore, the following review compares the literature not only with the original brief, but also with the updated industrial targets where relevant.

2.1.1 Pulse Amplitude

The pulse amplitude relates to the voltage and current magnitudes across the load during the active portion of each pulse cycle. In ECM, the applied voltage & current have a significant effect on the electrochemical dissolution rate. Under fixed conditions, a higher applied voltage produces a higher current density and, according to Faraday's law, this increases material removal rate (MRR) which is often taken as a performance parameter to reduce machining time without influencing the quality of the process [1] [7]. However, this relationship is complicated by the impedance of the inter-electrode gap (IEG), which varies during machining due to material dissolution and electrode feed [2]. Consequently,

delivering the required pulse magnitude across the load is one of the most critical performance requirements to achieve the correct current density for the ECM power supply.

The voltages reported in the literature vary considerably depending on the scale and application of the process. In conventional ECM, Kozak [8] reported voltages of 8 – 30 V applied between electrodes with a gap size of 0.2 – 0.8 mm and producing a current density of 10 – 400 A/cm². Similarly, Tang et al. [9] achieved a material removal rate of 411.4 mm³/min on S-03 stainless steel while applying a voltage of 24 V and achieving a current density of 10 – 180 A/cm². On the other hand, micro ECM power supplies operate at significantly lower voltages to improve the machining localization. For instance, Spieser and Ivanov [10] developed a PSU for μ ECM with a voltage pulse amplitude of 1 – 10 V and a current of up to 5 A: electrochemical turning and hole drilling were performed with pulses of –7 V and 8 V, respectively. More recently, Zhang et al. [11] proposed a bipolar PSU for micro ECM with positive pulses of 14 V and negative pulses of –5 V, used for machining tungsten carbide while maintaining an IEG value of 20 μ m.

Against this background, the updated TextureJet specification indicates a shift from extremely high voltage operation towards higher current capability. In particular, the revised current requirement of ± 20 A is broadly consistent with the research, while the revised voltage of ± 150 V is closer to, but nevertheless exceeds many reported machining voltages. The elevated requirement, following the relationship between voltage and IEG width:

$$J = \frac{\kappa(V - \Delta V)}{\delta}, \quad (2.1)$$

where κ is the electrolyte conductivity, V is the applied voltage, ΔV is the total overpotential, and δ is the IEG width, likely reflects TextureJet’s specific demand for a machining configuration with a higher-resistance. [8].

2.1.2 Pulse Waveform and Timing

The pulse waveform defines the temporal structure of the voltage applied across the ECM load. In a pulsed waveform, the timing characteristics of the pulses - such as frequency and duty cycle - are critical parameters. For bipolar systems, the polarity sequence and

the off-time between transitions must also be considered. These parameters are fundamental to the ECM process because they govern the electrochemical conditions within the inter-electrolyte gap.

Compared to a continuous DC current, a pulsed waveform introduces periodic off-times during which no current flows through the IEG. During these intervals, the electrolyte can eliminate the reaction products and gas bubbles from the gap, dissipate the Joule heating, and allow the electrical double layer (EDL) to discharge before the next pulse [6] [12]. In addition, studies [12] - [14] reported an enhanced electrolyte flow condition in the IEG, improved localization of anodic dissolution, and small and stable gaps found when utilizing a pulsed machining process, leading to greater process stability, better suitability for control, simpler tool design, and higher machining accuracy. The latter is strongly influenced by the pulse duration because of the EDL's charging dynamics. Schuster et al. [15] utilized ultrashort pulse voltages (< 100 ns) to control the charge time of the EDL, ultimately achieving an improved machining accuracy through a higher frequency and a narrower pulse width. However, the use of ultrashort pulses strongly reduces the efficiency of the ECM, because the pulse-off period, used to achieve a complete discharge of the EDL, must be several tens of times longer than the pulse-on period [16]. Therefore, studies [11] [16] [17] suggest the use of bipolar (polarity-reversing) pulses with frequencies ranging from 10 to 50 kHz, since such approach allows the EDL to be actively discharged by driving a reverse current; this significantly reduces the machining time by increasing the duty cycle of the process. Furthermore, Aliakseyeu [17] argues that pulses should follow the waveform $\underline{(+ve) V} \rightarrow \underline{(0) V} \rightarrow \underline{(-ve) V} \rightarrow \underline{(0) V} \rightarrow \underline{(+ve) V}$ with adjustable pulse on and off widths depending on the material to machine.

In the context of these findings and initial requirement, the updated TextureJet specification is more closely aligned with the literature, particularly in its emphasis on high frequency and short pulse widths. The requirement for frequencies of up to 330 kHz and minimum pulse widths of $0.5 \mu\text{s}$ places the design within a pulsed micro ECM regime reviewed above. Furthermore, the waveform specification presented in Section 1.1, adheres to the undertaken research.

2.2 Circuit Topologies for Bipolar Pulse Generation

The generation of fast bipolar voltage pulses represents a central requirement for modern micro-ECM systems. Several circuit architectures have been proposed in the literature for this purpose, making use of various topologies including H-bridge (half & full-bridge) inverters, Marx generators and its solid-state variants, and hybrid topologies that combine elements of both. The suitability of the circuits depends on the application's requirements in terms of voltage levels, switching frequency, waveform flexibility, and circuit complexity.

2.2.1 Improved Solid-State Marx Generator

Zhu et al. [18] designed a bipolar square wave pulse power supply for liquid sterilization applications based on an improved Marx generator combined with a solid-state switch. Their experimental prototype improves on the traditional Marx generator by optimizing the count of switching components: each stage n is assembled by 4 IGBTs and diodes, compared to the 6 IGBTs and diodes required by a traditional bipolar Marx circuit, resulting in a saving of $2n$ IGBTs and diodes for an n -stage design [18, fig. 1]. According to the theory provided by Shi et al. [19], the stage number governs the size of the output voltage: capacitors are charged in parallel and then discharged in series through controlled IGBT switching, which also enables the creation of a bipolar output. In addition, the IGBT switching is enabled by a drive board with a maximum operating frequency of 18 kHz. Consequently, the three-stage prototype achieved a ± 6 kV output at frequencies of 50–500 Hz with pulse widths of 2–20 μ s [18, p. 856]. In addition, a notable contribution of this work is the design of a discharge circuit, which reduces the tailing time when driving resistive–capacitive loads: through a special control sequence of the pulse waveform, the initial tailing time of the falling edge of 47 μ s was reduced to approximately 300 ns.

Taking into account the problem specification of this project, the maximum operating frequency of 500 Hz is several orders of magnitude below the maximum operating frequency of 330 kHz specified by TextureJet. Furthermore, the presented voltage output is significantly larger than the required voltage, but this can be adjusted by supplying the circuit with a smaller DC input voltage or decreasing the number of stages. The pulse width is broadly consistent with the requirements.

2.2.2 Boost Bipolar Pulse Generator (BBPG)

Chen et al. [20] proposed a bipolar pulse generator that combines a boost converter with a Marx-style discharge circuit, referred to as a boost bipolar pulse generator (BBPG). Their boost circuit operates independently of the bipolar pulse repetition frequency: the frequency of the boost switch S_B is 50 kHz, while the bipolar pulse frequency is 5 kHz [20, p. 215]. The input–output voltage relationship of the BBPG [20, eq. 3] is given by

$$U_o = \frac{n}{1-D} U_{in}, \quad (2.2)$$

where n is the number of stages and D is the boost duty cycle; the input voltage U_{in} is increased to an output voltage U_{out} through five operating modes per cycle: parallel capacitor charging, positive series discharge, a second parallel capacitor charging, negative series discharge, and a final charging interval. Additionally, the amplitude of U_{out} can be adjusted without altering the circuit’s stage count by changing the size of the duty cycle. A key contribution of this paper is the integration of a peak closed-loop control method to address pulse amplitude tuning under varying load conditions. The output pulse is attenuated through a resistance–capacitance voltage divider with a ratio of $k = 1000$, and the peak is sampled by a detection circuit based on the OPA350 models of operational amplifiers with a bandwidth of 16.07 kHz [20, p. 214]. The sampled peak is then fed back to a TMS320F28335 DSP controller, which performs incremental PID adjustment of the boost duty cycle to maintain constant pulse amplitude [20, p. 215]. The five-stage prototype demonstrated stable bipolar pulse output of ± 2 kV at a repetition frequency of 5 kHz with pulse widths of 5 to 10 μ s from a 100 V input. The boost circuit achieved an efficiency above 95 % and the overall system efficiency exceeded 80 % [20, Tab. II–III].

Against the specifications of TextureJet, the validated repetition frequency of 5 kHz remains well below the maximum operating frequency of 330 kHz, while the pulse width is broadly consistent with the requirements. The output voltage of ± 2 kV exceeds the ± 150 V requirement, however, as per Eq. 2.2., it is scalable through the duty cycle D and the number of stages n . Furthermore, the bandwidth of the peak detection circuit of 16.07 kHz would be insufficient for closed-loop control at higher frequencies.

2.2.3 Bipolar High-Voltage Pulsed Power Source (BHVPPS)

Malviya and Veerachary [21] proposed a boost converter-based bipolar high-voltage pulsed power source (BHVPPS) that, like Chen et al.'s BBPG [20], integrates a boost converter with a Marx-style series capacitor discharge. The n -Stage BHVPPS consists of an inductor, n number of capacitors, and $(4n + 2)$ switching devices [21, p. 5223] and operates through six distinct modes: positive pulse generation, dead time, negative pulse generation, dead time, capacitors charging from the inductor, and capacitors holding their voltages [21, pp. 5224-5225]. A distinguishing feature of this work is the proposed control scheme of the pulses, which allows the load voltage to appear during the inductor charging period, eliminating the need to wait for the inductor current to reach zero before delivering pulses. This provides a broader duty ratio range for the boost switch S_B and helps in achieving outputs with enhanced amplitudes [21, p. 5223]. The two-stage experimental prototype generated ± 500 V bipolar pulses at 5-6 kHz with pulse widths of 20-40 μ s from a 20-40 V input, achieving a voltage amplification of approximately 18 times for resistive loads and 16 times for resistive-capacitive loads [21, pp. 5229-5231]. The pulse rise time was measured to be close to 131 ns, which is approximately 1 % of the pulse width [21, Fig. 14]. The authors also provided a comparative analysis between existing topologies, demonstrating that the BHVPPS achieves a higher voltage gain while requiring a low-voltage input source [21, Tab. IV].

In comparison to the TextureJet specification — ± 150 V, ± 20 A, 330 kHz maximum frequency with 0.5 μ s minimum pulse width — the proposed BHVPPS topology offers a suitable foundation. The six-mode operating sequence produces the required bipolar waveform, and the comprehensive design procedure provided by the authors, such as capacitor selection, inductor sizing, and stage count choice, facilitates the precise adaptation of the circuit to the TextureJet operating requirements.

Chapter 3

Design of the PSU

3.1 Selected Topology

Based on the requirements established in Chapter 1 and the literature review carried out in Chapter 2, the bipolar high-voltage pulsed power source (BHVPPS) proposed by Malviya and Veerachary [21] was selected as the foundation topology for the present PSU. This choice is justified through the evaluation of the three reviewed power supplies against TextureJet’s specifications on four metrics: voltage adaptability, completeness of design methodology, compatibility with non-ideal loads, and the absence of major architecture-level barriers to higher-frequency implementation.

Unlike conventional Marx-derived circuits, the BHVPPS achieves high output voltages from a low DC supply, while being able to adjust the output’s amplitude through both the number of stages and the duty ratio of the boost switch S_B [21, p. 5223]. Malviya and Veerachary provide the most comprehensive design procedure of the three papers reviewed: the sizing expressions for the inductor, capacitors, and stage number are given [21, pp. 5227–5229], facilitating the adaptation of the circuit to TextureJet’s specifications. In addition, the BHVPPS was analysed and experimentally validated for both purely resistive and resistive-capacitive loads: the IEG presents a dynamic impedance that cannot be adequately represented by a purely resistive model. Finally, with respect to higher-frequency implementation, the BHVPPS presents fewer architecture-level constraints than the other reviewed candidates (Chen et al.’s peak-detection circuit is limited to 16.07 kHz, Zhu et al.’s IGBT driver operates at a maximum frequency of 18 kHz).

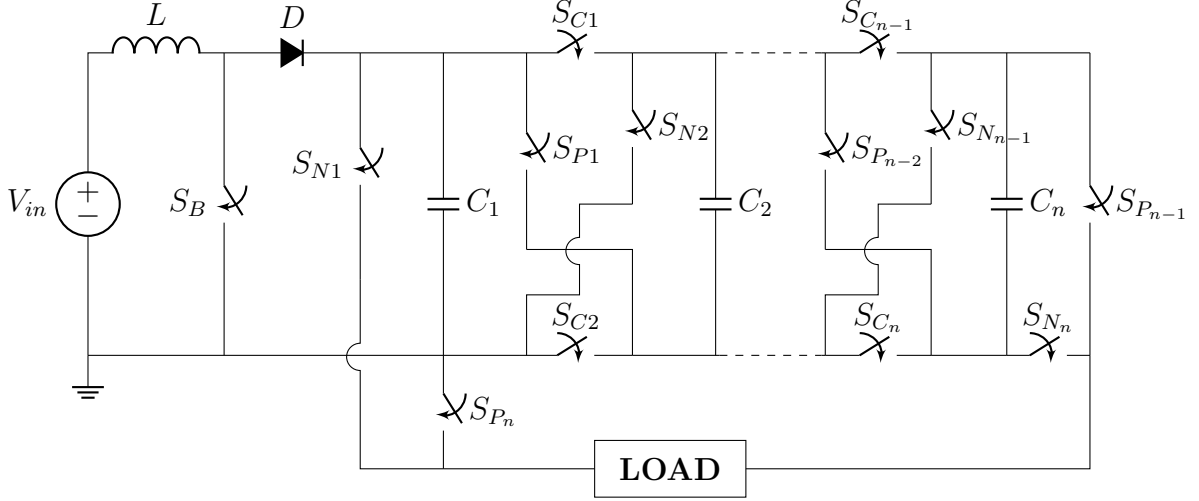


Figure 3.1. Circuit diagram of the n -stage BHVPPS, redrawn from [21].

3.1.1 Operating Principle

The BHVPPS combines a front-end boost stage with a Marx-type capacitor network to generate very high bipolar voltages from a comparatively low-voltage DC source. For an n -stage implementation, the circuit comprises an input inductor L , a diode D , n number of storage capacitors ($C_1, C_2, \dots, C_n$), a boost switch S_B , $(n + 1)$ positive pulse switches S_{P_i} , $(n + 1)$ negative pulse switches S_{N_i} , and $2(n - 1)$ capacitor-charging switches S_{C_i} , giving a total of $(4n + 2)$ switching devices [21, p. 5223].

The fundamental operating principle consists of two steps. First, each capacitor, while connected in parallel, is charged to the input voltage level through the charging network and inductor. Second, the capacitors are reconfigured in series through either the positive or the negative switch set, so the stored energy appears as a high-voltage pulse of the required polarity. Furthermore, the achievable pulse amplitude in the BHVPPS is influenced by both n and the duty ratio of the boost stage S_B , which provides an additional degree of freedom for amplitude control for the present design [21].

3.1.2 Configuration of Operation

The BHVPPS operates through six distinct modes within each period T_s , which collectively produce a complete bipolar pulse cycle. The “Type-B” control scheme is adopted over the earlier “Type-A” control scheme because it allows the load voltage to appear

during the inductor charging period, thereby eliminating the constraint that the inductor current must reach zero before pulse delivery [21] [22]. This provides a broader duty ratio range for S_B and enables the generation of pulses with enhanced amplitude and wider pulse widths [21, p. 5223]. The BHVPPS has the following six switching patterns and corresponding circuit states, as shown in Fig. 3.2(a)–(e).

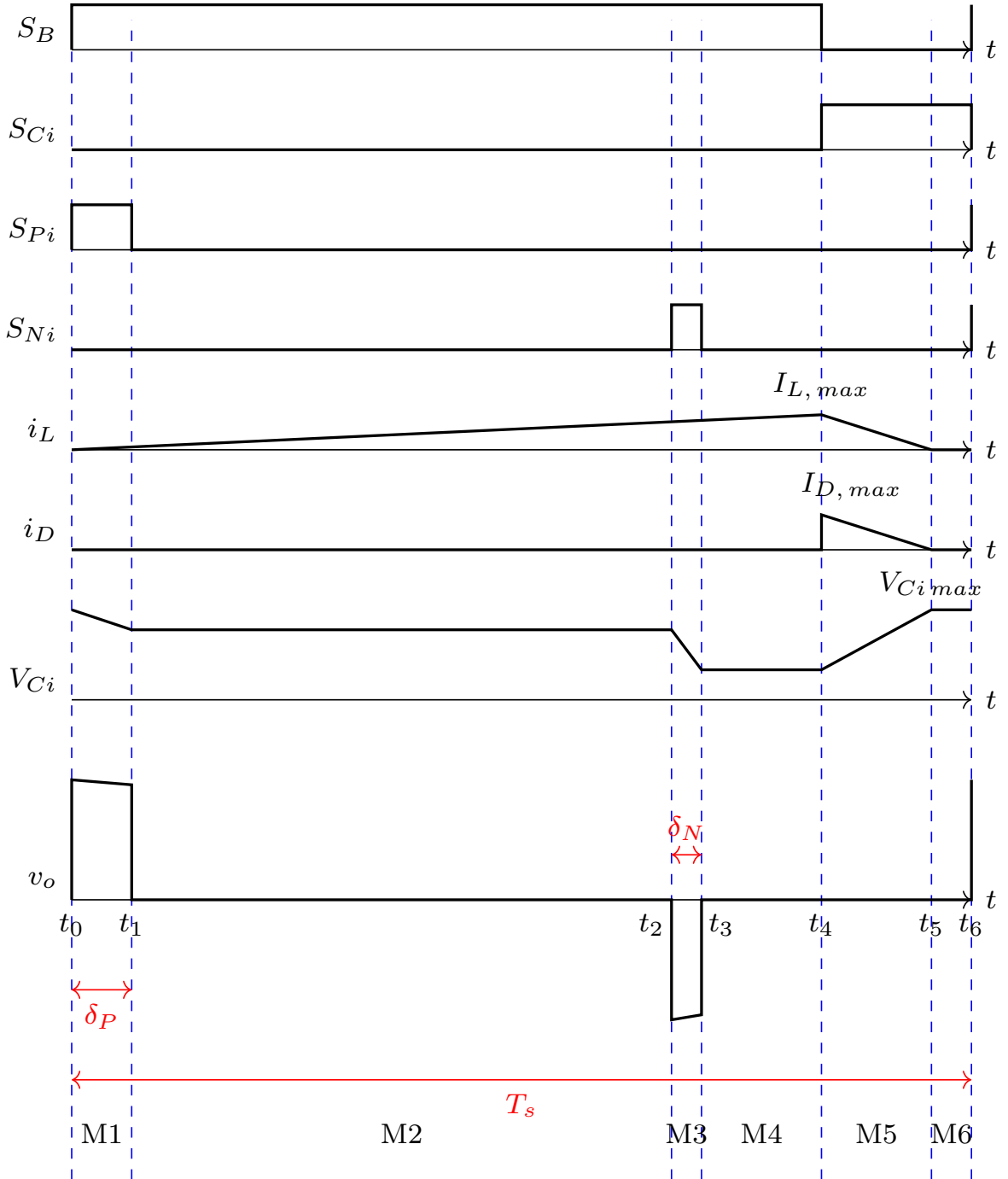


Figure 3.2. Idealized key-waveforms of BHVPPS (see Appendix B)

Mode 1: In this mode (M1), the boost switch S_B and all positive pulse switches S_{P_i} are turned on, while S_{N_i} , S_{C_i} , and diode D remain off. Therefore, the inductor L is connected to the input source v_{in} and stores energy, while the charged capacitors are placed in series, delivering the positive output pulse δ_P across the load.

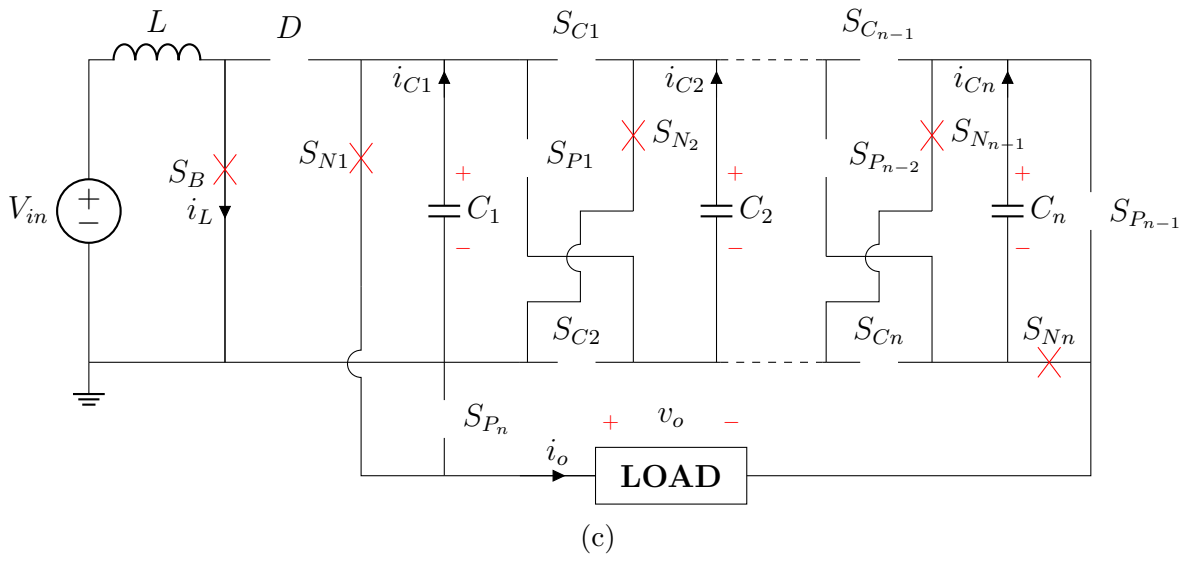
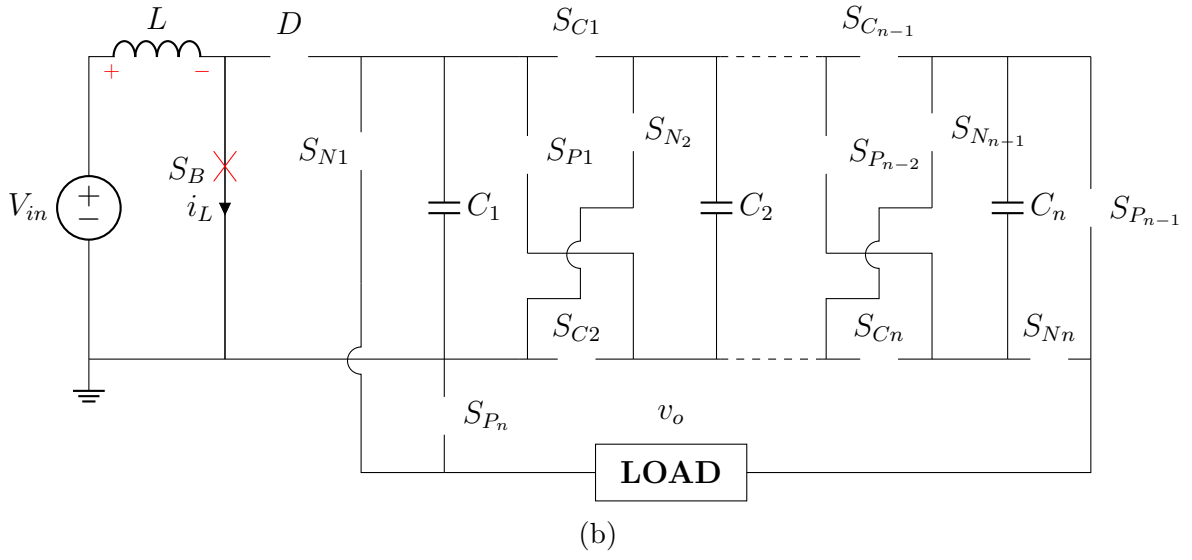
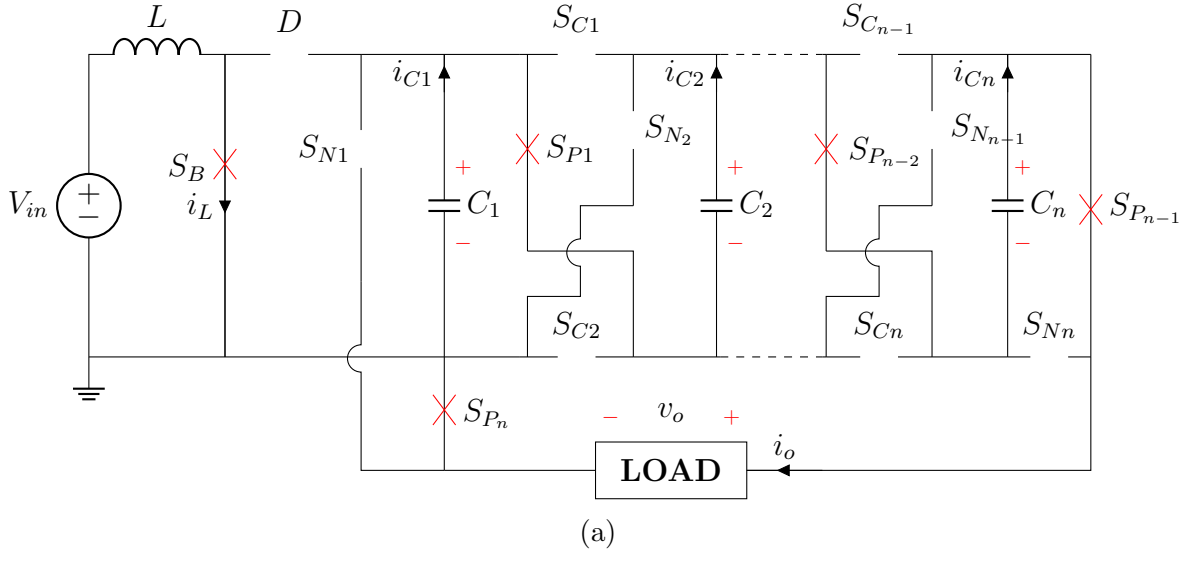
Mode 2: Mode 2 (M2) provides the interval between the pulses of positive and negative polarity. Only S_B remains in the conducting mode, therefore, the inductor continues charging from the source, while the capacitors retain their previous voltages levels and the load voltage is zero. This interval is particularly useful when a real-capacitive load is used, as it has to be discharged every time before the delivery of the next pulse.

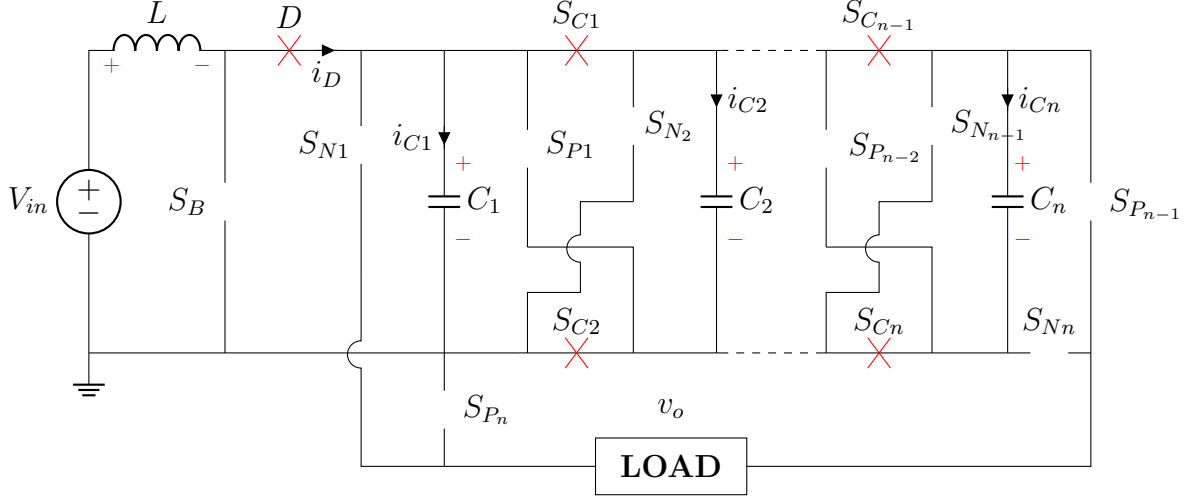
Mode 3: Mode 3 (M3) mirrors M1 but with reversed polarity. All negative pulse switches S_{N_i} are turned on, while S_{P_i} , S_{C_i} and diode D remain off. The capacitors are again connected in series with the load, now with the opposite orientation, so a negative pulse δ_N is produced across the load while the inductor continues to charge through S_B .

Mode 4: This mode (M4) is analogous to M2. Only S_B conducts, the inductor L continues to store energy, the capacitor voltages are held, and the load again sees zero voltage. The durations of Modes 2 and 4 can be adjusted according to the requirements.

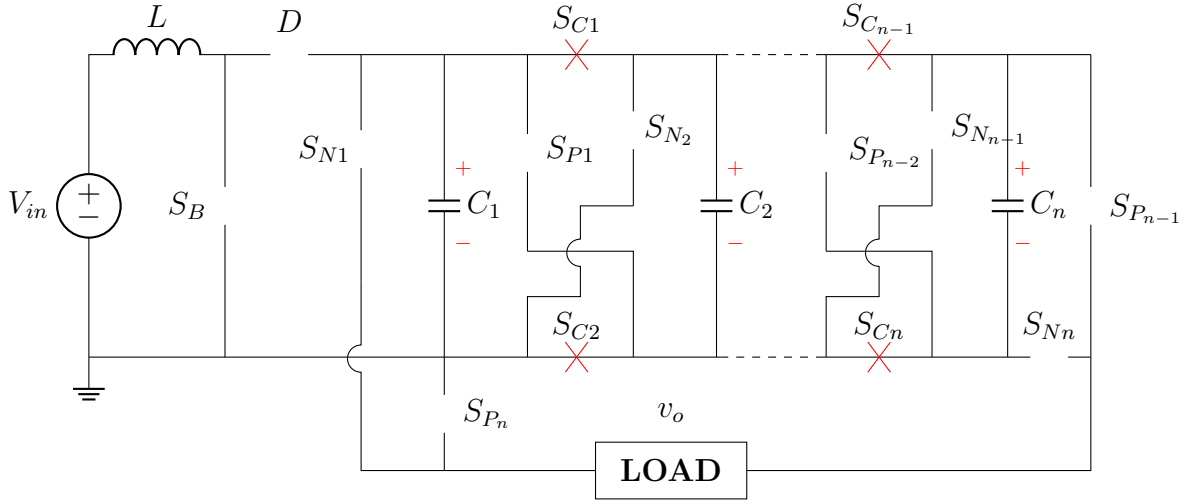
Mode 5: In Mode 5 (M5), S_B is turned off and the charging switches S_{C_i} are turned on, which forward-biases diode D . The energy stored in the inductor from M1 to M4 is equally distributed across the capacitors $C_1, C_2, \dots, C_n$ which are connected in parallel and charged to the same voltage level, v_C . Consequently, the load voltage during M1 and M3 is n times the voltage appearing across a single capacitor, v_C . Furthermore, the inductor current falls to zero by the end of this mode and the diode becomes reverse-biased.

Mode 6: Mode 6 (M6) lasts until the end of the switching cycle. Switches S_{P_i} and S_{N_i} and the boost switch S_B remain off, the charging switches S_{C_i} remain on, diode D stays reverse-biased, and the capacitors $C_1, C_2, \dots, C_n$ hold their voltages. No voltage appears across the load in this interval.





(d) Circuit D



(e) Circuit E

Figure 3.3. BHVPPS operations: (a) M1, (b) M2 and M4, (c) M3, (d) M4, (e) M5.

3.2 Component Design and Sizing

3.2.1 Capacitor Design

The selection of the intermediate capacitor value C is governed by the maximum allowable dip in pulse voltage during the active portion of each pulse. During M1 and M3, the stage capacitors are connected in series and discharge through the load resistance, causing the output voltage to decay from its peak value [23]. If C is too small, the resulting voltage droop becomes excessive and the output pulse no longer approximates the desired flat-top form [24]. Conversely, an excessively large C increases the stored energy that must be

replenished during M5, which extends the inductor discharge time and may conflict with the timing constraints at high switching frequencies [24]. Therefore, the capacitor design represents a tradeoff between pulse quality and the timing budget of the switching cycle.

1) *Resistive Load*: For a purely resistive load R , the output voltage decays exponentially from its peak value as [21, Eq. 15]:

$$v_o(t_1) = V_{max}e^{-nt_1/RC}, \quad (3.1)$$

where $V_{max} = nV_{C,max}$. Finding the fractional voltage dip at the end of a pulse of width δ and further rearranging of Eq. 3.1 yields the expressions:

$$\frac{\Delta V}{V_{max}} = 1 - e^{-n\delta/RC}, \quad (3.2)$$

$$C = \frac{n\delta}{R[\ln(V_{max}/v_o(\delta))]}, \quad (3.3)$$

where δ is the pulse width and $v_o(\delta)$ is the output voltage at the end of the pulse.

2) *Resistive-Capacitive Load*: The IEG presents a resistive-capacitive (R-C) impedance: the electrolyte path contributes a resistance R_L , while the EDL formed introduces a capacitance C_L [16]. Malviya and Veerachary [21] derive the R-C load expression:

$$v_o(t) = nqV_{C,max}(1 - e^{-A(t-t_0)}) + nV_{C,max}e^{-A(t-t_0)}, \quad (3.4)$$

where the decay constant A and the parameter q are defined as:

$$A = \frac{1}{CR_L}\left(\frac{C}{C_L} + n\right), \quad q = \frac{1}{C_LR_LA}. \quad (3.5)$$

Consequently, the capacitor value for a specified dip can be determined from Eq. 3.4 by rearranging and taking the natural logarithm:

$$\ln\left\{\frac{C}{nC_L}\left[\left(\frac{v_o(t_1)}{V_{max}} - 1\right) + \frac{nC_Lv_o(t_1)}{CV_{max}}\right]\right\}, \quad (3.6)$$

$$C = \frac{nC_L[v_C(t_1)]^2}{\{[v_C(t_0)]^2 - [v_C(t_1)]^2\}}. \quad (3.7)$$

Stage Count	Capacitor Count	Max. Stage Voltage (V)	Capacitor Size (μF)	Droop (%)
2	2	75	52	5
3	3	50	78	5
4	4	37.5	104	5
5	5	30	130	5

Table 3.1: Comparison of candidate stage numbers

The table above displays the required capacitor size for each candidate stage count, computed from Eq. 3.3 with a fixed voltage droop of 5 %. A higher stage count demands a proportionally larger capacitance to maintain the same pulse quality and the maximum stage voltage decreases with n , reducing the voltage stress on the switching devices, but increasing the total count of components. Therefore, a two-stage configuration ($n = 2$) is selected for the present design. This choice offers the lowest total capacitance requirement (52 μF), the fewest components (10 switching devices as derived in Section 3.1.1), and has been experimentally validated by Malviya and Veerachary [21] at this exact stage count. Furthermore, a maximum stage voltage of 75 V is well within the rating of commercially available MOSFETs, so there is no device-stress motivation to increase n beyond 2.

Having established the stage count, the load impedance is characterised to determine the required capacitance. For the purely resistive model, the load resistance is obtained directly from Ohm’s law as $R = \frac{V}{I} = 7.5 \Omega$. For the resistive-capacitive load, the same electrolyte resistance $R_L = 7.5 \Omega$ applies, with an additional EDL capacitance $C_L = 560 \mu\text{F}$ at the electrode–electrolyte interface [3][8][25].

With $n = 2$ and both load models established, the computed capacitance from Eq. 3.3 is 52 μF , which falls between the standard values of 47 μF and 56 μF . Selecting $C = 56 \mu\text{F}$ ensures a voltage droop $< 5 \%$ and allows the inductor current to discharge fast enough. Substituting back into Eq. 3.2 (resistive) and Eq. 3.4 (resistive–capacitive) to verify:

$$\frac{\Delta V}{V_{\max [R \text{ load}]} } = 4.650 \%, \quad \frac{\Delta V}{V_{\max [R-C \text{ load}]} } = 4.762 \%. \quad (3.8)$$

3.2.2 Inductor Design

The front-end inductor L serves as the primary energy-storage element in the BHVPPS and forms part of the front-end boost charging stage, with the purpose of raising the low input DC voltage [26]. During M1-M4, the inductor charges from the input DC source through the boost switch S_B , and in M5 this stored energy is transferred to the stage capacitors. The operation in discontinuous conduction mode is especially useful because the inductor current is driven back to approximately zero each switching cycle, which simplifies the cycle-by-cycle energy definition used later for inductance sizing [27]. For the present dissertation, the inductor is sized for the operating point presented in Appendix B, namely $v_o = \pm 150 \text{ V}$, $i_o = \pm 20 \text{ A}$, $\delta_P = 10 \mu\text{s}$, $\delta_N = 5 \mu\text{s}$, and $T_s = 150 \mu\text{s}$.

Under Type-B control, V_{in} is the voltage that appears across the inductor during M1-M4 and can be expressed in terms of the duty cycle [28], given by:

$$V_{in} = V_{C,max}(1 - D) = 12.5 \text{ V}, \quad (3.9)$$

where

$$D = t_4/T_s. \quad (3.10)$$

Assuming the inductor current starts at zero and rises linearly during M1-M4, the peak current at t_4 is given by [21, Eq. 4]:

$$I_{max} = \frac{V_{in}t_4}{L}. \quad (3.11)$$

The corresponding energy stored at this point is therefore [21, Eq.53]:

$$E_{in} = \frac{LI_{max}^2}{2} = \frac{V_{in}^2 t_4^2}{2L}. \quad (3.12)$$

Neglecting switching, conduction, and magnetic losses, the energy stored in the inductor over one switching cycle is equated to the energy delivered to the load during the bipolar output pulses; in other words, the law of conservation of energy states that $E_{in} = E_{out}$.

Hence, for the selected nominal pulse pair:

$$E_{out} = \int_0^{\delta_P} v(\delta_P)i(\delta_P) dt + \int_0^{\delta_N} v(\delta_N)i(\delta_N) dt = 0.030 + 0.015 = 0.045 \text{ J}. \quad (3.13)$$

Substituting this into Eq. (3.11) gives the required inductance:

$$L = \frac{V_{in}^2 t_4^2}{2E_{out}} = 27 \mu\text{H}, \quad (3.14)$$

However, the selection of the inductance L is also constrained by the discontinuous conduction mode (DCM). In practical terms, the inductor current must fall to zero before the end of M5. In other words, the discharge interval must satisfy $t_5 < T_s$. Using the critical-inductance expression derived by Malviya and Veerachary [21, Eq. 43–45]:

$$L < \left\{ \frac{1}{\omega I_{max}} [(v_c(t_4) - V_{in}) \tan(\omega(T - t_4))] \right\}, \quad (3.15)$$

where

$$\omega = \frac{1}{\sqrt{nLC}} \quad (3.16)$$

is the frequency of the L-C network during M5. With the previously selected values $n = 2$ and $C = 56 \mu\text{F}$, the DCM verification becomes the summary shown below.

Parameter	Value
$v_C(t_4)$	69.83 V
I_{max}	57.87 A
ω	18.185 rad s ⁻¹
V_{in}	12.5 V
L_{crit}	27.41 μH
L	27 μH
$L < L_{crit}?$	Yes — DCM not violated

Table 3.2: DCM verification for $L = 27 \mu\text{H}$

Since the selected inductance remains below the critical value, this subsection closes with a clear design conclusion: the chosen inductor satisfies both the cycle-by-cycle energy requirement and the DCM constraint, given the adopted stage count and capacitor value.

3.2.3 Switching Device Selection

The selection of the active switching devices is a critical design decision that directly affects the efficiency, switching performance, and overall feasibility of the bipolar PSU. The two principal candidates for solid-state switching in power electronic converters of this class are the Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) and the Insulated-Gate Bipolar Transistor (IGBT) [29]. The choice between them is governed primarily by the voltage and current stress imposed on the devices, the required switching speed, and the suitability of each device for pulsed operation with short duty cycles [30].

In the present design, the maximum voltage across any individual switching device should not exceed 75 V. The peak inductor current, as derived from the steady-state analysis, reaches approximately 57.9 A. The on-state resistance of a power MOSFET is dominated by the drift region resistance, which increases sharply with the device breakdown voltage rating due to the need for a thicker and more lightly doped epitaxial layer [29]. At voltage ratings < 200 V, power MOSFETs present on-state resistances in the low-m Ω range, with low conduction losses that scale quadratically with the device current [31]:

$$P_{MOSFET} = I_{rms}^2 R_{DS(on)}. \quad (3.17)$$

In contrast, the IGBT exhibits a collector–emitter saturation voltage ranging from 1.5 V to 3.0 V, arising from the intrinsic p–n junction drop at the collector side [32]. At high voltages > 400 V and high currents > 40 A, this would result in substantially lower conduction losses than those of a MOSFET. Such a conduction advantage of the IGBT over the MOSFET occurs at high powers due to the rise of the MOSFET’s drift region resistance and the IGBT’s conductivity modulation property [30], [33]. Therefore, while MOSFETs suffer from a high R_{DS} , IGBTs benefit from a low and almost constant $V_{CE(sat)}$:

$$P_{IGBT} = V_{CE(sat)} I_C D \quad (3.18)$$

Equally important in the present application is the switching behaviour of the devices, which must be able to operate at 6.667 kHz and drive pulse widths of 10 μ s for the pos-

itive output phase and $5\text{ }\mu\text{s}$ for the negative output phase. MOSFETs, being unipolar devices, do not have a minority carrier storage time during switching and can achieve turn-on and turn-off transitions in the range of tens of nanoseconds [1]. Conversely, an IGBT is a minority carrier device and the movement of electrons in the bipolar portion produces a characteristic tail current that can persist for several hundred nanoseconds to several microseconds depending on the device technology and operating temperature [32]. In the BHVPPS, where the shortest active interval is only $5\text{ }\mu\text{s}$, such tail currents could potentially increase turn-off losses and overlap with subsequent switching states [34].

A practical consideration that arises from the use of MOSFETs in this topology is the presence of the intrinsic body diode. This parasitic p–n junction diode is a structural feature of every conventional silicon power MOSFET, formed between the p-type body region and the n-type drain drift layer [29] [35]. In standard half-bridge or full-bridge configurations, the body diode serves a useful function by providing a path for inductive current during dead-time intervals [36], while in the present converter certain switch positions may allow the body diode to conduct even when the MOSFET channel is intentionally switched off, hence, creating unintended current paths. Such parasitic conduction paths, sometimes referred to in the literature as sneak circuits, have been identified as a general concern in power converters operating multiple switching states [37]. In the context of the present BHVPPS, these unintended paths may result in uncontrolled discharge of the stage capacitors, cross-conduction between stages, or distortion of the output waveform.

Despite the low device voltage stress and the fast-switching advantage associated with MOSFETs, the IGBT is selected for the present BHVPPS design because reverse conduction behavior is a more critical constraint in this topology than absolute switching speed. Additionally, the IGBT would present lower conduction losses while operating close to a current of 57.9 A , however, it could introduce higher switching losses and is less favorable for very short pulses due to minority-carrier tail current. For these reason, the IGBT offers a more robust device choice for the present topology.

Chapter 4

Results and Evaluation

4.1 Simulation

A time-domain LTspice model of the selected two-stage BHVPPS was developed to verify the analytical configuration presented in Chapter 3. The resistive-load case was evaluated first, as it provides the clearest validation of the converter operating sequence. The simulated circuit used the design values established earlier in the dissertation which are summarized in Table 4.1. The switching devices were modeled as ideal voltage-controlled switches in order to isolate the topology behaviour from device-specific non-idealities (switching losses, tail currents, and parasitic capacitances) and because of the unavailability of IGBT models for the LTSpice software. The simulation was performed for 5 ms, with a total period of 150 μs and pulse widths of $\delta_P = 10 \mu\text{s}$ and $\delta_N = 5 \mu\text{s}$.

Parameter/ Component	Value/ Part no.	Parameter/ Component	Value/ Part no.
V_{in}	12.5 V	C	56 μF / 890764429002CS
V_o	150 V	L	27 μH / AGP4233-153
$f_{(pulserate)}$	6.667 kHz	Switches	IKQ50N120CT2
δ_P	10 μs	Diode	VS-E5PH6012
δ_N	5 μs	R	7.5 Ω
R_L	7.5 Ω	C_L	560 μF

Table 4.1: Parameters Used in the LTspice Model

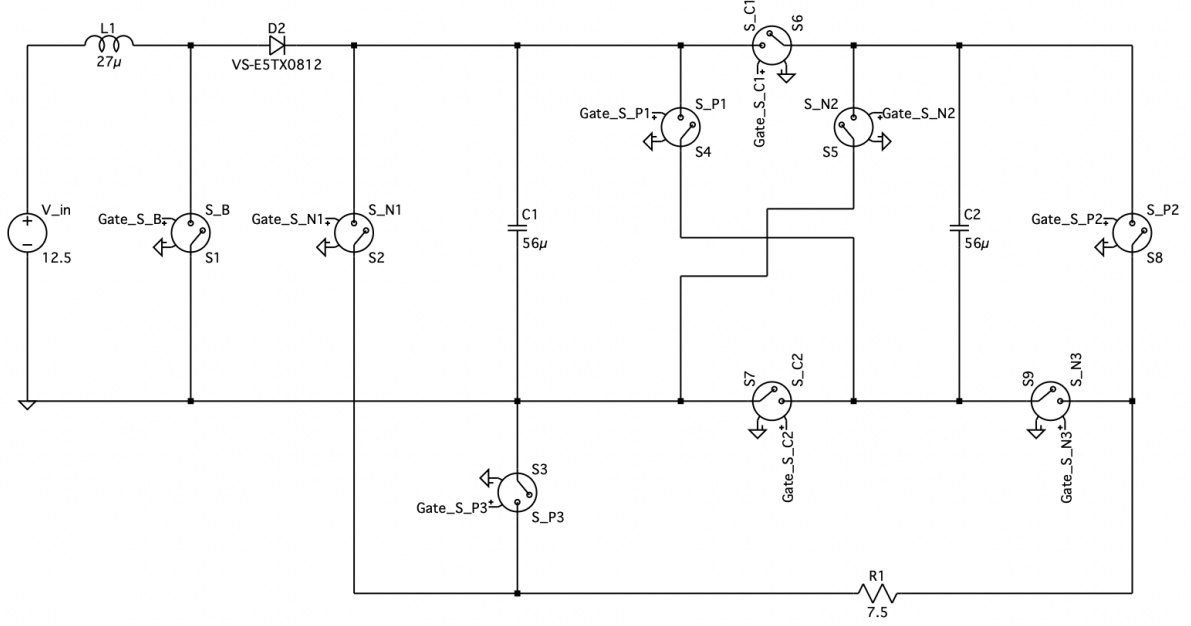


Figure 4.1. Complete two-stage BHVPPS circuit constructed in LTSpice

4.1.1 Gate Drive Signals

Fig. 4.1 shows the four gate drive waveforms that implement the six-mode Type-B control scheme over two complete switching periods ($T_s = 150 \mu\text{s}$). Positive pulse switches S_{Pi} are enabled for $\delta_P = 10 \mu\text{s}$ at the start of each period (M1), while negative pulse switches S_{Ni} are enabled for $\delta_N = 5 \mu\text{s}$ during M3. The boost switch S_B operates during M1-4, with a delay of $100 \mu\text{s}$ at the start of each cycle, during which the inductor charges from the input source; afterward, capacitor-charging switches S_{Ci} turn on during Mode 5, after S_B is turned off, to transfer the stored inductor energy to the stage capacitors. Although the switching devices are modelled as ideal, the gate drive signals were configured with a rise and fall time of 1 ns each to approximate a realistic gate transition. More importantly, the on-time of each gate signal was reduced by 4 ns relative to the ideal timing to introduce a small dead-time between complementary switch groups. This precaution ensures that selected switch sets are not conducting simultaneously and in a physical implementation, this is essential to prevent shoot-through currents.

The timing sequence observed in the simulation is consistent with the idealised key waveforms shown in Fig. 3.2 and confirms that the control logic, presented with greater detail in Appendix C, correctly produces all six operating modes within each switching cycle.

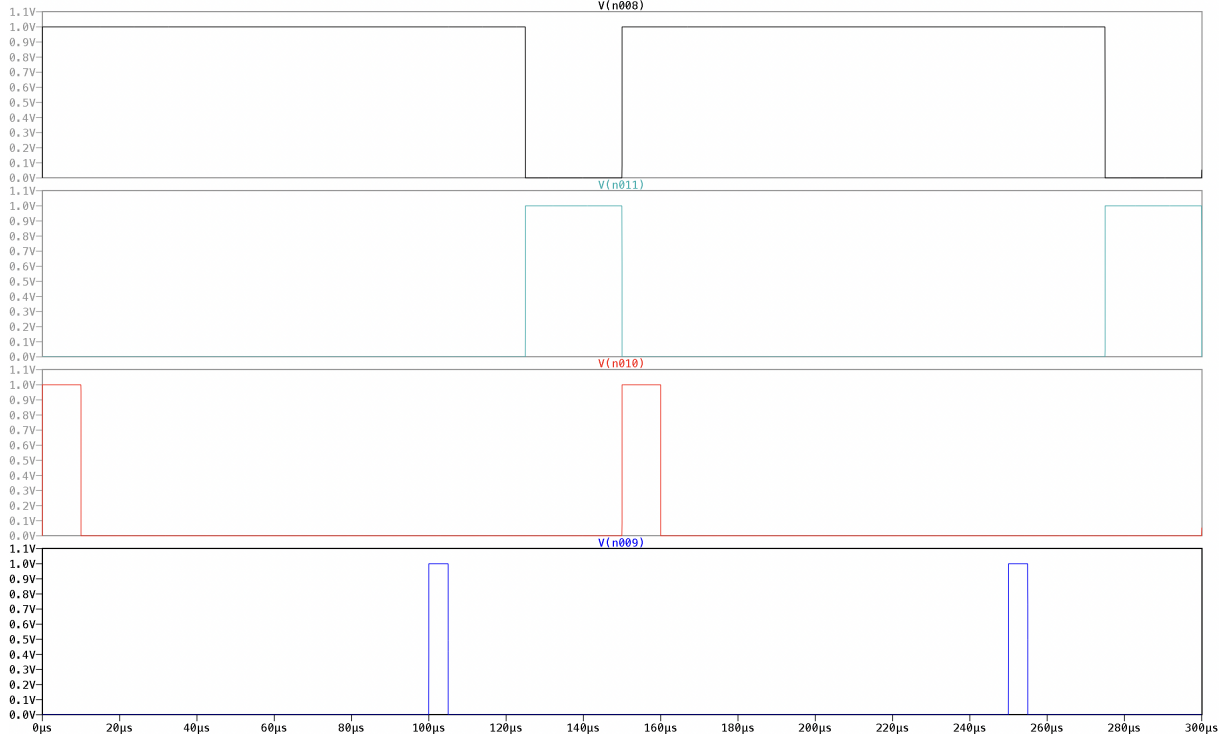


Figure 4.2. Simulated gate drive signals over two switching periods: S_B (top), S_{Ci} (second), S_{Pi} (third), and S_{Ni} (bottom).

4.1.2 Output Voltage & Current Waveform

The simulation was performed over a 5 ms time window and produced a voltage of 34 V and a current of 4.5 A throughout the resistive load, as depicted in Fig. 4.3. The waveform exhibits an initial transient period during which the pulse amplitude is marginally stable and undergoes an oscillation, as the stage capacitors accumulate energy (see Appendix D). Nonetheless, the six-mode switching sequence produces the specified bipolar pulse sequence: $(+34) \text{ V} \rightarrow (0) \text{ V} \rightarrow (-31) \text{ V} \rightarrow (0) \text{ V} \rightarrow (+34) \text{ V}$ waveform. The slight asymmetry in the positive and negative pulse amplitudes is likely attributable to the unequal pulse widths ($\delta_P = 10 \mu\text{s}$, $\delta_N = 5 \mu\text{s}$), as a longer positive pulse draws more charge from the stage capacitors, producing a higher output voltage.

As expected from Ohm's law, the current mirrors the voltage waveform in shape and timing, ranging from approximately +4.5 A to approximately -4.0 A. The current returns cleanly to zero during the dead-time intervals (M2 and M4), confirming that no residual conduction paths exist through the ideal switch network when all pulse switches are open.

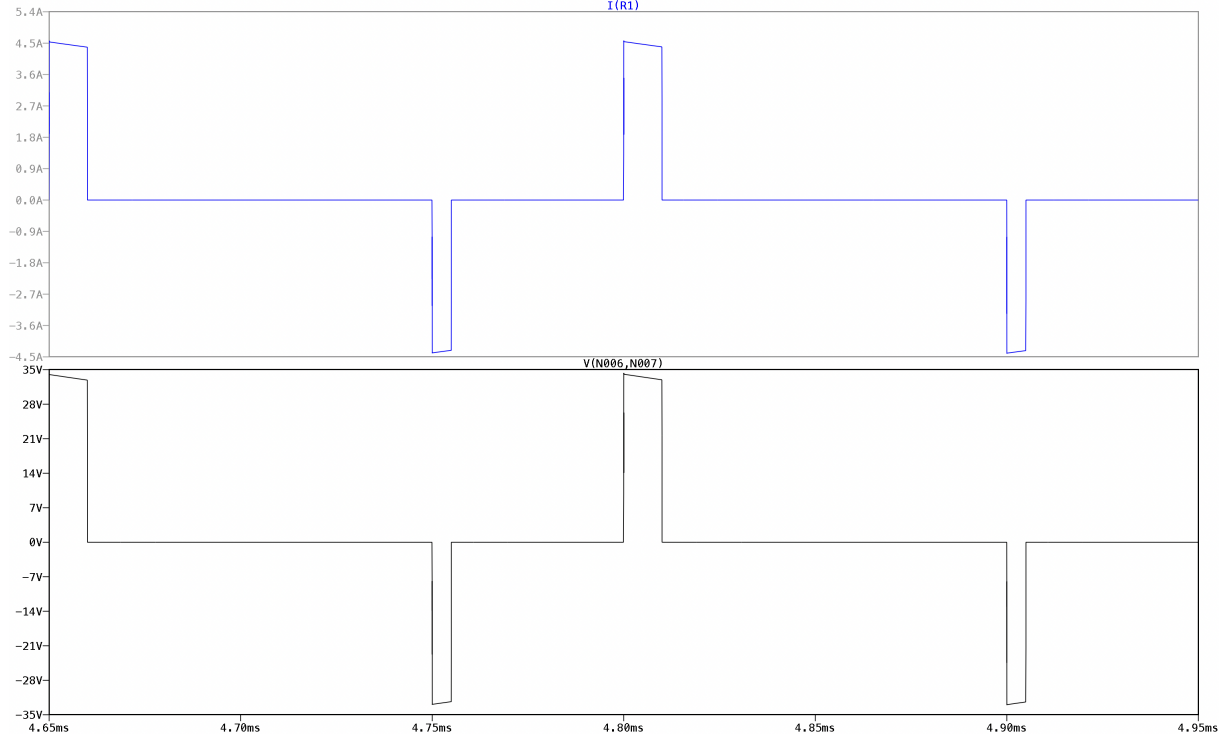


Figure 4.3. Magnified view of the simulated output voltage & current at the steady state operation across the $7.5\ \Omega$ resistive load.

4.1.3 Voltage Droop Analysis

A key design requirement established in Section 3.2.1 was that the pulse voltage droop must remain below 5 % to maintain a flat-top pulse suitable for ECM operation. Fig. 4.4 shows a magnified view of a single positive pulse at steady state, and the voltage decays from a peak of approximately 34.3 V to approximately 32.9 V over a pulse duration of 10 μ s, yielding a measured voltage droop of:

$$\left. \frac{\Delta V}{V_{\max}} \right|_{\text{sim}} = \frac{34.3 - 32.9}{34.3} \times 100 \approx 4.08\ \%. \quad (4.1)$$

This is below the 5 % design ceiling and is broadly consistent with the analytical prediction of 4.65 % from Eq. 3.8. Therefore, the measured result confirms that the selected capacitance of 56 μ F satisfies the pulse-quality requirement. However, the small difference between the simulated and analytical droop values can be attributed to the difference in capacitor voltage compared to the theoretical calculations, since the latter are larger.

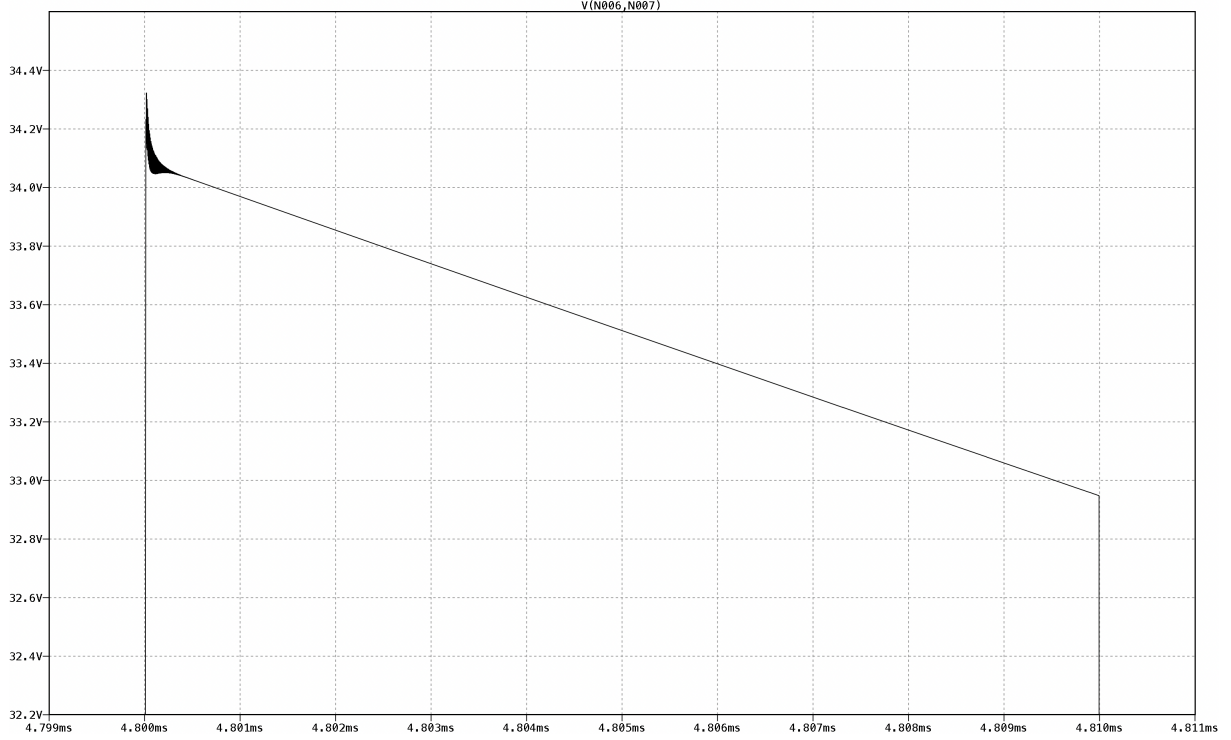


Figure 4.4. Magnified view of a single positive output pulse showing a droop of $\approx 4.08\%$.

4.1.4 Limitations and Recommendations for Future Work

The steady-state output values of approximately 34 V and 4.5 A are notably lower (4.4 times) than the theoretical design targets. The simulation indicates that each capacitor charges to approximately 16.5 V, suggesting that the steady-state energy balance between inductor charging and capacitor discharge has not been fully achieved. Additionally, the simulation presented was limited to a purely resistive load and an attempt to model the resistive-capacitive load ($R_L = 7.5\ \Omega$, $C_L = 560\ \mu\text{F}$), was not fully successful (see Appendix E), as the introduction of the load capacitance induced numerous oscillations.

Further work is required to resolve the amplitude shortfall. Possibly, systematically varying the input voltage and duty cycle could help find the target capacitor voltage of 75 V. With respect to the resistive-capacitive load, the oscillatory behaviour may be mitigated through the introduction of snubber circuits or damping networks across the output, or by revising the capacitor sizing to shift the resonant frequency away from the pulse bandwidth. Finally, replacing the ideal switches with manufacturer IGBT models would enable quantification of conduction and switching losses, tail-current effects, and their impact on pulse quality at the target operating point.

Chapter 5

Project Plan and Timeline

5.1 Management Plan

The project was carried out throughout a 24 week period, and tracked against an initial Gantt Chart. A comparison between the planned and actual timings is presented below:

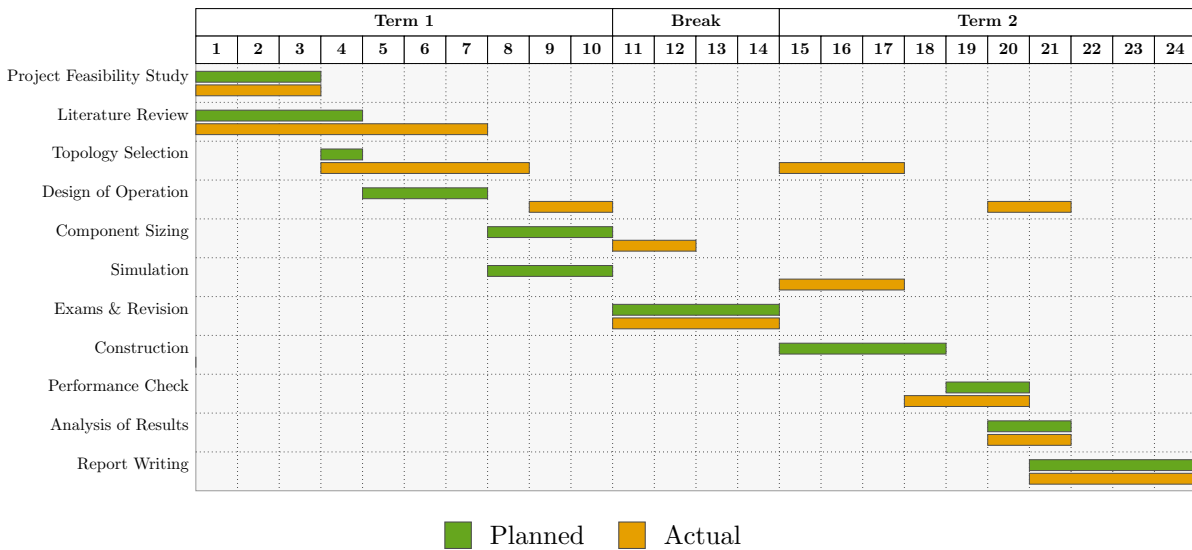


Figure 5.1. Comparison of planned & actual progress across the 24-week project schedule.

1. The **Literature Review** was extended by three weeks due to the breadth of content related to the bipolar pulse generation.
2. The **Topology Selection** and **Design of Operation** were each split across Terms 1 and 2, as initial analyses required revision after deeper understanding of the switching modes.

3. The **Construction** was not initiated due to the increased difficulty of the project compared to initial expectations.
4. The **Report Writing** was compressed from four to three weeks as a result of the accumulated delays.

5.2 Risks and mitigation

Likelihood and impact were assessed qualitatively as Low (1), Medium (2), or High (3). The overall risk level was determined from the combined severity of likelihood and impact.

Risk Description	Likelihood	Impact	Risk Level	Mitigation Measures
Increased circuit complexity beyond initial expectations.	High	High	9	Validation of each design step independently before proceeding, and testing of different device values.
Simplified modelling neglects parasitic effects or switching non-idealities.	High	High	9	Consider possibilities through literature, and perform multiple rounds of careful testing.
Simulation model failing to reproduce theoretical waveforms.	Medium	High	6	Incremental model construction, allowing to verify each operating mode individually.
Analytical miscalculations in component sizing propagate to simulation.	Medium	Medium	4	Performing independent verification of all sizing equations against the source literature.
Repetitive strain or fatigue from prolonged computer-based work.	Medium	Medium	4	Regular breaks and tasks varied between analytical, simulation, and writing work.
Loss of project data or simulation files.	Low	High	3	All project files kept on a cloud-synced repository, such as Overleaf.

Table 5.1: Project risk assessment and mitigation measures

5.3 Budget

This section summarizes the estimated cost of the experimental prototype. The main power stage components include two 56 μF capacitors (890764429002CS, $\sim\text{£}8$ each), one 27 μH inductor (AGP4233-153, $\sim\text{£}5$), 9 IKQ50N120CT2 IGBT switches ($\sim\text{£}10$ each), one VS-E5PH6012 diode ($\sim\text{£}4$), and one 12.5 V battery ($\sim\text{£}20$), bringing the component subtotal to approximately $\text{£}135$ [38] [39]. Additional costs may arise, such as the PCB fabrication, gate driver ICs, connectors, heatsinks, and others.

5.4 Ethical Awareness

Safety: The BHVPPS is designed to deliver high voltage outputs with peak inductor currents approaching 58 A. At these levels, contact with energised conductors poses a serious risk of electric shock, burns, and, in extreme cases, fatal injury [40]. While the present work remained at the simulation stage, any future hardware implementation must incorporate proper insulation, over-current and over-voltage protection, and clearly marked high-voltage warning labels in accordance with IEC-61010 [40].

Environmental Impact: A responsible deployment of the PSU must be accompanied by appropriate disposal procedures in compliance with local environmental regulations. On the other hand, ECM itself offers environmental advantages as the absence of mechanical cutting forces eliminates tool wear debris and the non-thermal nature of the process avoids the generation of hazardous fumes or thermally altered surface layers [2].

Intellectual Property: The design specifications used throughout this project were provided by TextureJet Ltd and all proprietary information has been used solely for the purposes of this academic project. The BHVPPS topology itself is drawn from the published work of Malviya and Veerachary [21]. No commercial exploitation of the design has been undertaken and would require the explicit consent of TextureJet and the University.

Integrity of Reporting: Throughout this dissertation, simulation results have been reported transparently, including the significant discrepancy between the achieved steady-state output and the design target, as well as the unsuccessful resistive-capacitive load simulation. No data have been selectively omitted or adjusted to present a more favourable outcome. This commitment is consistent with the Code of Practice on Research Integrity.

Chapter 6

Conclusion

This dissertation presented the design and simulation-based validation of a two-stage bipolar high-voltage pulsed power source (BHVPPS) for ECM, following the specifications provided by TextureJet Ltd. A critical literature review confirmed that the target specifications are broadly consistent with the reported ECM parameters, and the BHVPPS topology proposed by Malviya and Veerachary [21] was selected on the basis of its comprehensive design methodology, high voltage gain from a low-voltage input, and compatibility with resistive-capacitive loads.

With the use of the LTspice platform, the simulation has yielded the correct six-mode operation and a voltage droop of approximately 4.08 % (within the design ceiling). However, the steady-state output reached a reduced output in amplitude, roughly 4.4 times below the design targets, and the resistive-capacitive load simulation exhibited severe oscillations. The hardware construction phase was not initiated due to the increased complexity encountered during the design of the PSU. Despite these shortfalls, the project delivers a complete analytical design framework that can serve as a foundation for future hardware realisation. Recommended next steps include sweeping the input voltage and duty cycle to reach the target capacitor voltage, introducing networks to suppress the R-C load oscillations, and replacing the ideal switch models with manufacturer IGBT device models to quantify real-world losses.

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Appendix A: Comparison between Preliminary and Updated TextureJet Specifications

This appendix summarizes the evolution of the TextureJet design brief from the preliminary specification provided at the start of the project to the later, more precise industrial requirements. The updated values should be interpreted as capability limits rather than fixed continuous operating conditions. In particular, several updated values represent maximum achievable limits, while the final design point selected in this dissertation is intended to operate close to, but not necessarily at, those limits.

Parameter provided	Preliminary specification	Updated specification
Voltage capability	0 to ± 500 V	0 to ± 150 V
Current capability	0 to ± 6 A	0 to ± 20 A
Frequency	166 to 250 Hz	Max. 330 kHz
Waveform type	Bipolar square wave	Bipolar square wave
Min. pulse width	Not explicitly stated	$0.5 \mu s$

Table A.1: Comparison of preliminary and updated TextureJet specifications

Appendix B: Provided Waveform by TextureJet

Figure B.1 shows a screenshot of the BAXEDM control interface provided by TextureJet Ltd., which is used to configure and monitor the bipolar pulsed ECM process in real time. In the configuration shown, the positive and negative pulsed intervals are set to $10\text{ }\mu\text{s}$ and $5\text{ }\mu\text{s}$ respectively, with a total period of $T_s = 150\text{ }\mu\text{s}$ and bipolar pulse duty cycle of 10 %. The waveform display at the top of the interface confirms the expected bipolar current pattern: a positive pulse of $+20\text{ A}$ followed by a zero-current interval, a negative pulse of -20 A , and a second zero-current interval, consistent with the $(+ve)\text{ V} \rightarrow (0)\text{ V} \rightarrow (-ve)\text{ V} \rightarrow (0)\text{ V} \rightarrow (+ve)\text{ V}$ waveform specified in Section 1.1.

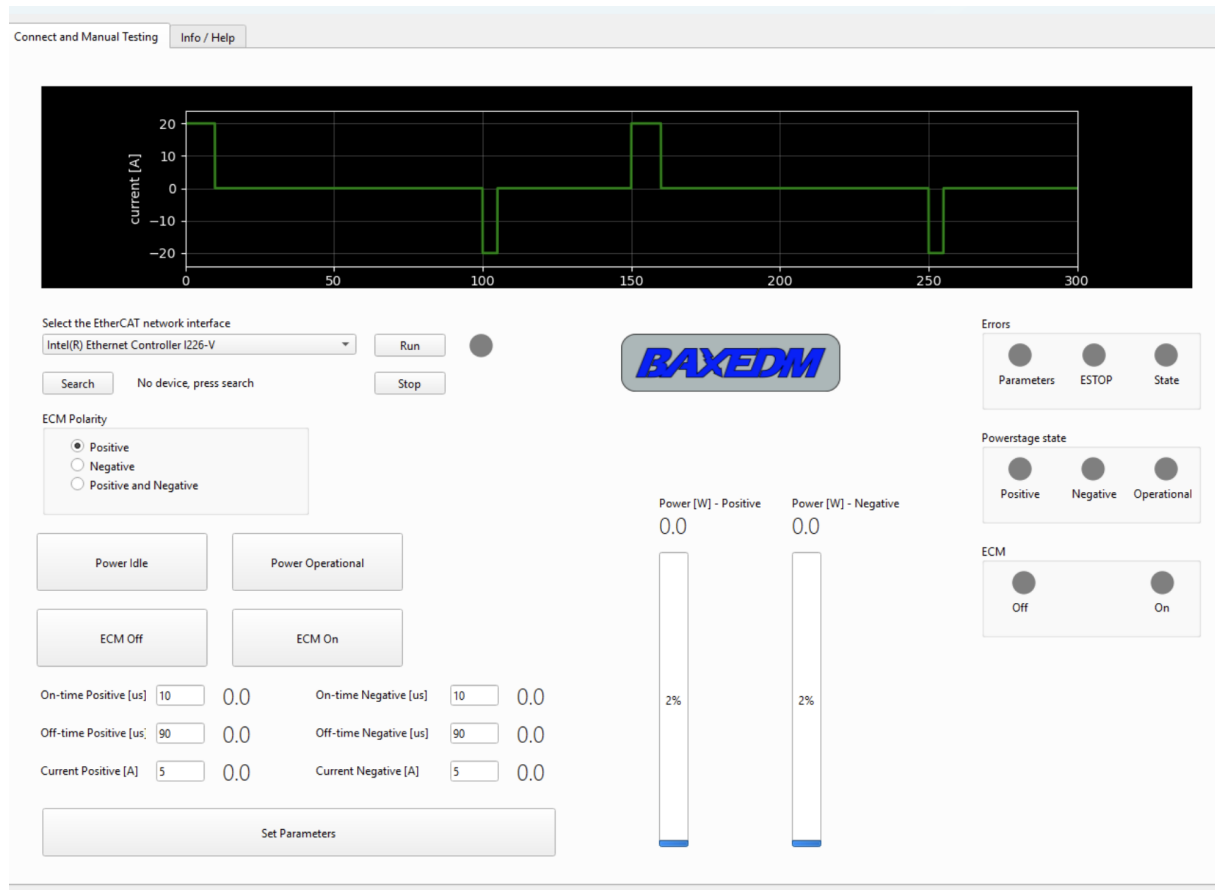


Figure B.1. Screenshot of the BAXEDM bipolar pulse parameter configuration and real-time current waveform display.

Appendix C: Control Logic – Ideal Switches

Each switching device is driven by an independent PULSE voltage source (V1–V9) connected through a $10\ \Omega$ series resistor to the corresponding gate terminal. The parameters of the pulse define the on-voltage, off-voltage, delay, on-time, period, rise time, and fall time. The switch models, listed on the right-hand side of the schematic, define the parameters for the ideal voltage-controlled switches: $R_{on} = 1\ \Omega$, $R_{off} = 1\ \text{M}\Omega$, and a threshold voltage of $V_T = 0.5\ \text{V}$.

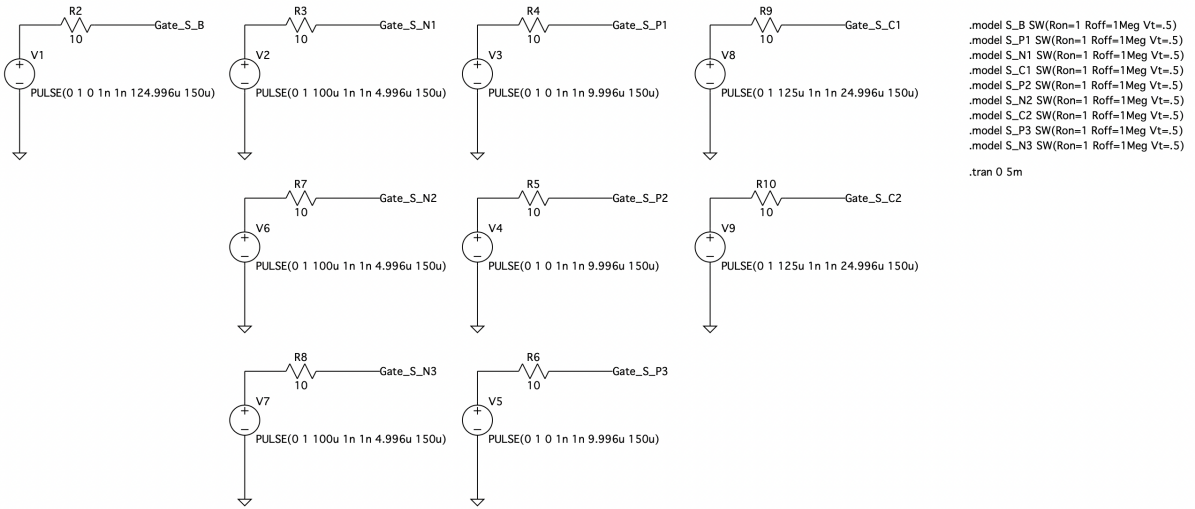


Figure C.1. LTspice gate drive circuit used to generate the control signals for all ten switching devices in the two-stage BHVPPS.

Appendix D: Output Current and Voltage (Resistive)

The figure below shows the load current and voltage in a simulation of 5 ms. Both outputs arrive at steady state after an initial transient charging of approximately 3 ms. The proportional relationship between the two traces confirms the purely resistive load behaviour ($i_o = v_o/R$) and verifies that no unintended phase shift or waveform distortion is introduced by the switching sequence.

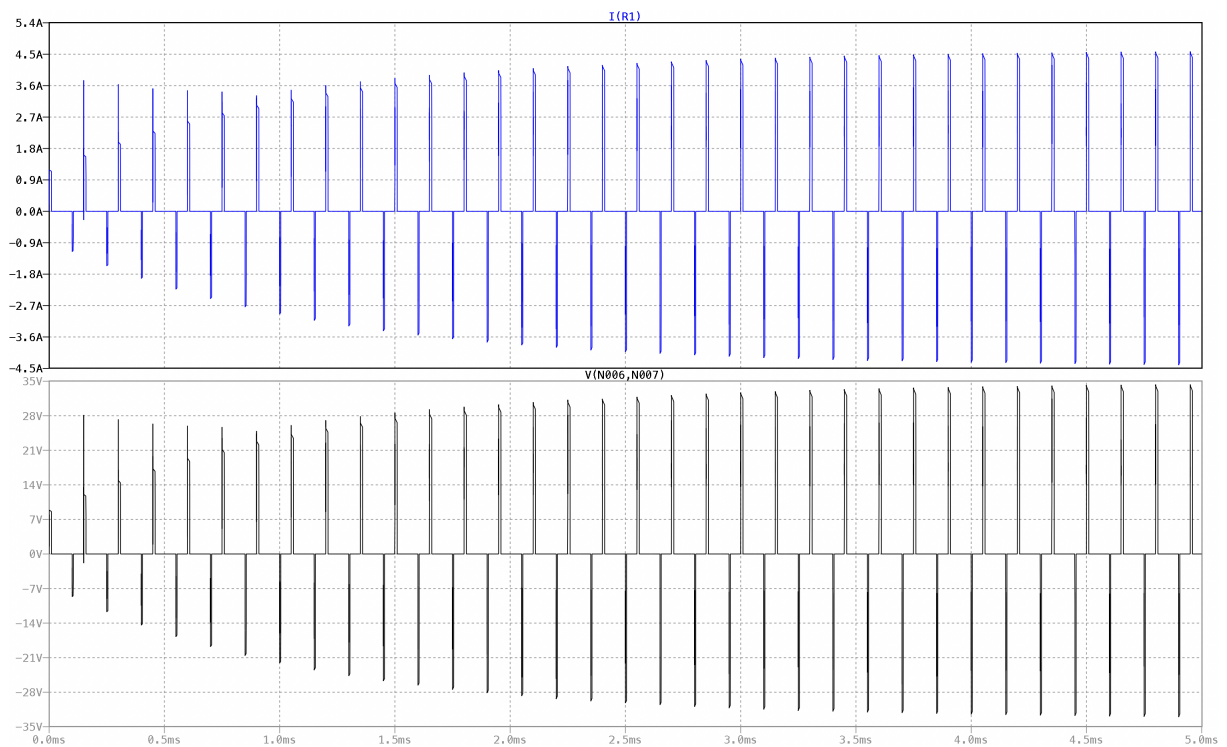


Figure D.1. Combined view of the simulated load current (top) and output voltage (bottom) across the resistive load over the full 5 ms transient simulation.

Appendix E: Output Current and Voltage (Resistive–Capacitive)

The output voltage and current obtained from a resistive–capacitive model exhibits large-amplitude oscillations, resulting in significant distortion of the intended bipolar waveform. Possibly, these oscillations are attributed to the resonant interaction between the stage capacitors and the load capacitance during the series connection.

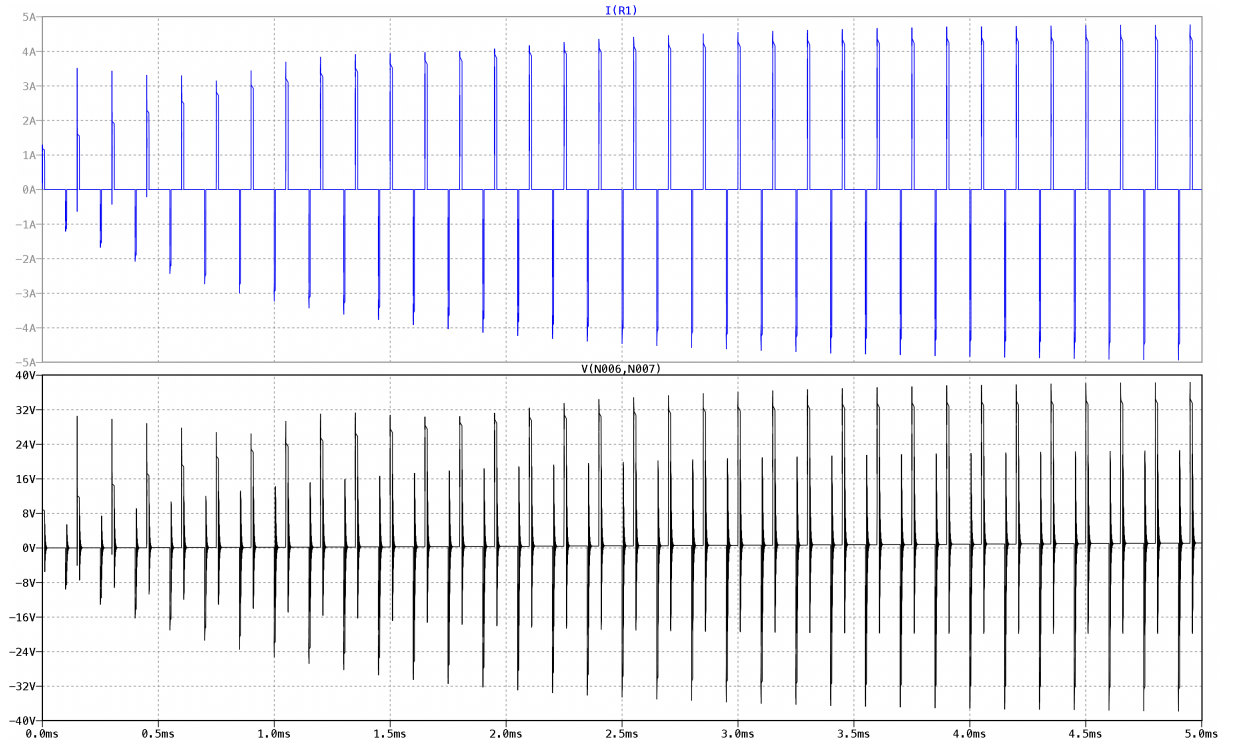


Figure E.1. Simulated output voltage across the resistive-capacitive load ($R_L = 7.5\Omega$, $C_L = 560\mu\text{F}$), showing severe oscillatory behaviour that distorts the intended square-wave pulse shape.